[METRIC]

A-A-59338A
April 14, 1999
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

WATER QUALITY ANALYSIS SET: PURIFICATION (WQAS - P)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This CID covers the performance requirements for the Water Quality Analysis Set: Purification (WQAS - P). The WQAS - P is intended to be used for the assessment of raw water quality at potential water points for water purification, in operations to monitor the treatment process, and to perform chemical analysis in assessment of the quality of potable water in support of the water purification mission.

2. SALIENT CHARACTERISTICS

2.1 Description. The Water Quality Analysis Set: Purification, WQAS – P, (hereinafter referred to as the set) shall consist of a case containing various equipment to determine the acceptability of water for treatment, treatment process monitoring, and for treated water potability. The set shall contain those commercial items, integrated together, necessary to meet all performance requirements, various support items common for water quality measurement, and one item of Government-furnished property (Water Testing Kit, Chemical Agent, M272). Since the set is to be an integration of various commercial items, some minor modifications may be necessary to meet all requirements. As a result, the set may be subject to test of these modifications such as weight test, accessibility test, drop test, watertight test, and instructions test.

2.2 Materials and components. The materials and components selected for use in the set are the responsibility of the contractor, with the exception of Water Testing Kit, Chemical Agent, M272, to be provided by the Government. The materials and components shall be of sufficient durability to meet the operational, environmental, and support item requirements specified in this document. All material that comes in contact with water samples shall conform to the applicable federal regulations for use with potable water. The materials shall not present any safety hazards to operating and maintenance personnel when used for its intended purpose. The contents,

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pieces, and parts incorporated into the set may be newly fabricated from recovered/recycled material to the maximum extent practical, provided the set meets all requirements of this specification. Used, rebuilt, or remanufactured contents, pieces, and parts shall not be incorporated in the set.

2.2.1 Deterioration prevention and control. The set shall be fabricated from compatible materials, inherently corrosion and deterioration resistant, or treated to provide protection against the various forms of corrosion and deterioration to which the material is susceptible.

2.2.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

2.2.3 Degradable materials. Materials or adhesives used in construction of the protective case or internal packaging material shall not deteriorate, dissolve, suffer surface separation, or deform after prolonged exposure to water, either by submersion, or storage in a high humidity environment.

2.2.4 Exterior materials. The case, in closed condition, may be exposed to chemical or biological contamination and shall be capable of being decontaminated without damage or degradation of the case integrity, or its contents. The outer material of the case shall be resistant to chemical and biological agents and to decontaminates used to neutralize these agents. The case, when subjected to agent or decontamination material application, shall not deform, dissolve, or leak. The exterior of the case and any closures, seals, or handles shall be designed so that any contamination remaining on, desorbed, or reaerosolized from the surface following decontamination shall not result in more than a negligible risk to unprotected personnel working within one meter. Material producer, case producer, and contractor certifications that the latter condition listed in this paragraph is met shall be acceptable.

2.2.5 Packaging material. The contents shall be secured in the case by a packaging material that will allow normal handling, transportation, and occasional drops, not greater than 1.3 meters, without damage to the contents. A means shall be provided to protect any electromagnetic pulse sensitive instruments while in the closed case.

2.3 Environment.

2.3.1 Operating environment. The set shall be capable of operating in an environment of 0 °C to 45 °C.

2.3.2 Storage environment. The set shall be capable of being stored at temperatures of 0 °C to 50 °C without damage to the case or the contents. The set should be capable of being stored at temperatures of -33 °C to 71 °C without damage to the case or the contents. Any chemicals which have a limited shelf life shall be labeled with the expiration date if unopened and the remaining life when opened, if applicable. Labeling shall be in accordance with ANSI Z 129.1.

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2.4 Protective case. The case shall be watertight. The case, with fully supplied contents, shall not exceed 13.6 kilograms in weight and 0.05664 cubic meters in volume. Case color and markings shall be as specified in the contract or order (see 6.1).

2.4.1 Pressure equalization. The case shall be equipped with a method to equalize the inside pressure with the outside pressure when closed.

2.4.2 Portability. The design of the case shall incorporate suitable means for carrying or manually transporting the case by a single person using one hand.

2.4.3 Loading diagram. A loading diagram showing content placement shall be permanently secured to the inside of the case. The loading diagram shall be on a stable material or otherwise treated to ensure water resistance and durability.

2.4.4 Accessibility. Contents shall be removable from and replaceable to the case by personnel wearing Mission Oriented Protective Posture (MOPP) IV gear (see 6.6) without damage to the packaging material or the contents of the case.

2.5 Performance.

2.5.1 Time limit. The total time required for performing a complete set of calibrations and tests (turbidity, pH/temperature, residual chlorine, and conductivity), and removal/replacement of components, GFP item M272 excluded, by a person wearing MOPP IV (see 6.6) gear shall be twenty (20) minutes or less.

2.5.2 Turbidity. The set shall measure the turbidity of various liquids, including fresh-water and salt-water, nephelometric turbidity units (NTU) preferred. The set shall have the following minimum performance characteristics for turbidity, as listed in Table I:

Table I: Turbidity Performance

a. Measurement range	0 to 150 NTU ± 10% relative accuracy; resolution of 0.1 NTU from 1 to 50 NTU; 1 NTU resolution from 50+ NTU.
b. Operating temperature	0 °C to 45 °C

2.5.2.1 Turbidity units. Sets that do not measure directly in NTU units shall include an appropriate conversion table from the measured units to NTU. Any required conversion table shall be permanently secured to the case and on a stable material or otherwise treated to ensure water resistance and durability.

2.5.2.2 Turbidity standards. Turbidity standards, as necessary for calibration and testing, shall be provided to include enough material to conduct a minimum of six (6) complete tests with calibrations, taking into account the possibility of standard contamination. Turbidity standards, which are reusable and resistant to contamination, shall be included when available.

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2.5.3 pH / Temperature. The set shall measure pH and temperature simultaneously, with pH measurements automatically compensated for temperature. The set shall present pH and temperature, although not necessarily simultaneously. The set shall have the following minimum performance characteristics for pH/Temperature, as listed in Table II:

Table II: pH/Temperature Performance

a. Measurement range, pH	2.00 to 12.00 with 0.01 resolution, relative accuracy of ±0.1.
b. Measurement range, temperature	-5 °C to 71 °C with 0.1 °C resolution, relative accuracy of ±1.0 °C.
c. Operating temperatures	0 °C to 45 °C

2.5.3.1 pH durability. Any pH-measuring device, which is submerged into the sample being tested, shall be capable of being dropped from 0.5 meters to a firm in-situ surface without damage, and should not be susceptible to desiccation.

2.5.3.2 pH standards. Any pH standards as necessary for calibration and testing shall be provided to include enough material to conduct at least twenty (20) complete calibrations, taking into account the possibility of standard contamination. The standards shall be capable of withstanding the storage temperatures cited herein without harm. An accuracy of ± 0.05 pH units shall be required. Standards requiring the use of raw, unprocessed water shall be buffered as necessary to prevent variations in pH due to water composition.

2.5.4 Residual chlorine. The set shall measure free residual chlorine without the visual judgment of the user. In addition, a color comparator shall be included to determine the residual chlorine content of water by a visual comparison between color standards of known value and the sample color. The color comparator shall require no power. The set shall have the following minimum performance characteristics for residual chlorine, as listed in Table III:

Table III: Residual Chlorine Performance

a. Without visual, residual chlorine	0 to 5.0 mg/L, 0.01 mg/L resolution, relative accuracy of ±0.05 mg/L.
b. Comparator, residual chlorine	0 to 10.0 mg/L.
c. Operating temperatures	0 °C to 45 °C

2.5.4.1 Chlorine test material. Any material required for use in determining free residual chlorine in water samples shall be provided. The material selected shall be stable during extended storage periods, not require measurement prior to use, and give accurate and repeatable results. Material shall be provided to conduct at least fifty (50) tests.

2.5.4.2 Chlorine standards. Any chlorine standards as necessary for calibration and testing shall be provided to include enough material to conduct at least twenty (20) complete calibrations, taking into account the possibility of standard

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contamination. Chlorine standards, which are reusable and resistant to contamination, shall be included when available.

2.5.5 Conductivity. The set shall measure total dissolved solids indirectly by measuring conductivity in water. Multiple methods required to cover the measurement range shall be clearly marked for the range each method covers. The set shall have the following minimum performance characteristics for conductivity, as listed in Table IV:

Table IV: Conductivity Performance

a. Measurement range	0 to 100 mS/cm, 0.1 µS resolution, ±1% relative accuracy.
b. Operating temperatures	0 to 45 °C

2.5.5.1 Conductivity units conversion. The set shall include a chart with the appropriate empirical conversions for fresh, salt, and product water over the measurement range to Total Dissolved Solids (TDS) units in mg/L. The chart shall be permanently secured to the case and on a stable material or otherwise treated to ensure water resistance and durability.

2.5.5.2 Conductivity standards. Any conductivity standards as necessary for calibration and testing shall be provided to include enough material to conduct at least twenty (20) complete calibrations, taking into account the possibility of standard contamination.

2.5.6 Support items. The following listed items provide support for the WQAS - P measurement tests.

2.5.6.1 Demineralizer. The set shall be capable of reducing the mineral composition of raw water to less than 0.4 ppm minerals. It shall be capable of demineralizing small quantities of water, approximately 170 grams (6 ounces) at a time. The container shall be unbreakable with a sealing top and a pouring spout. The demineralizing agent shall be replaceable and shall have a means of indicating when it is no longer effective.

2.5.6.2 Container. A break-resistant container of at least 150-milliliter capacity with permanently marked gradations in at least 25 ml intervals, and a pour spout, shall be supplied for sampling and holding liquids during operation.

2.5.6.3 Power supply. Any electrical devices shall be capable of operating on a self-sufficient electrical source, such as batteries, and should be capable of operating on standard 115 VAC, 60 Hz, single-phase power. When present, each device(s) unique alternating current (AC) adapter shall be marked to indicate which device(s) it supports. One for use and one spare for each self-sufficient power supply used by each component of the set shall be provided. Each component shall be capable of operating with its self-sufficient power supply for a minimum of fifty (50) complete calibrations and tests before maintenance.

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2.5.6.4 Hardware. Any tools, as necessary to conduct field maintenance, calibration, test procedures and exchange of self-sufficient power supply shall be provided.

2.5.6.5 Disposable towels. The set shall contain towels which are absorbent, low linting, and suitable for wiping small amounts of water from critical surfaces. Each towel shall have a surface area of at least 200 square centimeters and shall be contained in a dispenser that reduces airborne lint and keeps contamination out of the dispenser. The towels shall be disposable and biodegradable. A minimum of 250 towels shall be included.

2.5.6.6 Writing device. At least two fine-tip writing devices shall be provided. The devices shall contain permanent black ink that will not fade in the presence of moisture.

2.6 Government-furnished property (GFP). The Government shall furnish one M272 Chemical Agent Water Testing Kit (NSN 6665-01-134-0885) per set. The testing kit shall have interface dimensions of 25.1 cm x 15.9 cm x 7.0 cm, weigh no more than 1.22 kg, and shall fit within and be an integral component of the set.

2.7 Instructions. Abbreviated instructions on how to calibrate and how to perform the tests for Turbidity, pH/Temperature, Residual Chlorine, and Conductivity shall be permanently secured to the inside of the case to prevent loss. The instructions shall be on a stable material or otherwise treated to ensure water resistance and durability.

2.8 Identification Markings. The set shall be identified and marked in accordance with MIL-STD-130 to include, at a minimum, CID number, design activity code, manufacturer and CAGE, item nomenclature, serial number, NSN, weight, contract number, date of manufacture, technical manual number, and U.S. military property identification.

3. REGULATORY REQUIREMENTS

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. QUALITY ASSURANCE

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such compliance.

4.2 Inspection requirements. Bid samples, when required for CID items to ensure product quality, shall be as specified in the contract or order.

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4.3 Additional inspection. Additional inspections to verify modifications and integration of components shall be as required in the contract or order, to include any or all of the following: weight test, accessibility test, drop test, watertight integrity test, and instructions test.

4.3.1 Test conditions. Testing shall be conducted in ambient conditions with a temperature of 0 °C to 45 °C.

4.3.1.1 Weight test. The set, with all contents installed, shall be placed on a scale capable of weighing in 0.1-kilogram increments and the weight determined to the nearest 0.1 kilogram. The scale shall be in current calibration traceable to a national standard. A measured weight 0.1 kilograms or greater than the specified 13.6 kilograms shall be considered a failure.

4.3.1.2 Accessibility test. Testing personnel in MOPP IV gloves (see 6.6) shall remove the contents from the case, and upon completion, the contents shall be installed in the case per the loading instructions and the case closed and secured. Failure to remove all contents from the case, damaged caused to the case or packaging material, or inability to repack the case using the loading diagram shall be considered a failure of the accessibility test.

4.3.1.3 Drop test. The case, with contents, shall be dropped from a height of 1.3 meters onto an in-situ earth surface. The case shall be opened and the contents checked for visible damage. The drop test shall be repeated a second time and the examination repeated. Subsequent failure of any component to operate or function correctly because of damage shall be considered a failure of the drop test.

4.3.1.4 Watertight integrity test. The set, with contents and in a closed condition, shall be subjected to water spray through a nozzle directed to cover the parting line between any sections of the container and any external apparatus. The spray nozzle shall give uniform water distribution and shall have a flow capacity of not less than 4 liters/minute at 207-276 kPa water-line pressure, and water temperature between 4°C to 24°C. Fogging nozzles shall not be used. The spray shall be angled between 40° and 90° to the surfaces being sprayed. Spray time along each container surface shall not be less than 3 minutes continuous. The case shall be examined for deformation, dissolution, and separation of surfaces or evidence of any water leakage into the case. Deformation, dissolution, separation of surfaces or evidence of any water leakage into the case shall constitute failure of this test

4.3.1.5 Instructions test. The abbreviated instructions shall be used at least once to perform the Turbidity, pH/Temperature, Residual Chlorine, and Conductivity calibrations and measurement. Failure of the instructions to allow satisfactory completion of the test and obtain accurate results shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this publication.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 6.2).
c. Time frame required for submission of bid samples, and any additional inspections required (see 4.2 & 4.3).
d. Color of case and markings required (see 2.9).
e. The use of MOPP IV gear for content removal and arrangements to provide the necessary items (see 4.3.1.2).
f. When and where Government-furnished property will be available (see 2.6).
g. Packaging required (see 5.1).

6.2 Referenced Documents. The documents referenced in this CID shall be the issues in effect on the date of issuance of the invitation for bid or request for proposals unless otherwise specified. The documents form a part of this CID to the extent specified. In the event of a conflict between this CID and the documents referenced herein, this CID shall take preference.

6.2.1 Federal documents.

FED-STD-313	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

6.2.2 Department of Defense documents.

MIL-STD-130	Identification Marking of U.S. Military Property

Unless otherwise indicated, copies of federal and military standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.2.3 Non-Government publications.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z 129.1	Hazardous Industrial Chemicals - Precautionary Labeling

Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.

6.3 Government-furnished property. The contracting officer must arrange to furnish the property specified in 2.6.

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6.4 Cautions and warnings. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313; and 29 CFR 1910.1200 requires that the Material Safety Data Sheet for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the Material Safety Data Sheet(s).

6.5 Bid samples. When a bid sample inspection is required, the bid sample(s) must be identical to the production model. The contracting officer should include specific instructions in acquisitions documents regarding arrangements for examinations, approval of the bid sample results and disposition of the bid sample.

6.6 Definitions.

6.6.1 CAGE. An acronym for Commercial and Government Entity code.

6.6.2 CID. An acronym for Commercial Item Description.

6.6.3 GFP. An acronym for Government-furnished property.

6.6.4 Mission Oriented Protective Posture IV gear (MOPP IV). Military clothing to protect personnel from direct exposure to chemical or biological agents. Includes pants, jacket, gloves, foot coverings, and mask that allow personnel to continue normal activities in a hazardous environment.

6.6.5 NSN. An acronym for National Stock Number.

6.6.6 NTU. Nephelometric Turbidity Units; a commonly used standard of measurement for turbidity (clearness) of liquids.

6.6.7 Recovered materials. Recovered materials are those materials that have been collected from solid waste and reprocessed/recycled to become a source of raw materials, as distinguished from virgin raw materials.

6.6.8 Set. Reference to the Water Quality Analysis Set – Purification.

6.7 Data requirements. The requirement for technical manuals should be considered when this document is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12 L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including special tools), and instructions on shipment of the WQAS - P.

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6.9 Possible sources of products. Manufacturers of products known to meet some or all of the requirements of this CID are listed below. However, this list is only a small fraction of possible sources, is by no means exclusive, and competition is not limited to these companies.

Chemetrics, Inc.
Route 28
Calverton, VA 20138
Attn: Rob Mooney
www.chemetrics.com

Denver Instruments Company
6542 Fig Street
Arvada, CO 80004
Attn: Brian Taylor
www.denverinstrument.com

Hach Company
P.O. Box 389
Loveland, CA 80539
Attn: Brian Kindness
Attn: Dave Ruane
www.hach.com

Hanna Instruments, Inc.
Highland Industrial Park-584 E. Park Drive
Woonsocket, Rhode Island 02895-0849
www.hannainst.com

Marine Electric Systems
2 Bridewell Place
Clifton, NJ 07014-1135
Attn: Tony Gentile

O.I. Analytical
P.O. Box 9010
College Station, TX 77842-9010
Attn: Karl Brinkman
www.oico.com

Orion Research
500 Cummings Cntr
Beverly, MA 01915
Attn: Jane Stevens

Pelican Products, Inc.
23215 Early Ave.
Torrance, CA 90505
www.pelican.com

Pine Environmental Services, Inc.
379 Princeton Hightstown Rd.
Cranbury, NJ 08512
Attn: Jamie Gosweiler
www.pine-environmental.com

Specialized Products Company
1100 S. Kimball Ave.
Southlake, TX 76092-9009
www.specializedproducts.com

Sterling IPC
68950 Powell Road
P.O. Box 604
Romeo, MI 48065
Attn: Dan Lacy

YSI, Inc.
1725 Brannum Lane
Yellow Springs, OH 45387
Attn: Larry Norris
www.ysi.com

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6.10 Subject term (key word) listing.

Chlorine

Conductivity

pH

TDS

Total dissolved solids

Turbidity

Custodian
Army – AT
Navy – MS
Air Force – 99
GSA – FCGC

Preparing activity
Army - AT

Review activities
Army – EA, QM
DSMB - MB

(Project 6630-0577)

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